

Introduction to Python Programming

Welcome to the Introduction to Python Programming lecture. In this session, we will cover the fundamental concepts of Python programming, including the definition of code, syntax, and the role of the Python interpreter. By the end of this lecture, you will be able to write simple expressions and understand the output generated by the interpreter.

What is Code?

Code, or source code, is the sequence of instructions that make up a program. It is written in a programming language, which provides a set of rules and syntax that must be followed to ensure the code can be understood and executed by a computer.

Understanding Syntax

Syntax refers to the set of legal structures and commands that can be used in a particular programming language. In Python, syntax is crucial because it dictates how we write our code. For example, every statement must end with a newline, and indentation is used to define blocks of code.

The Role of the Python Interpreter

Python is an interpreted language, which means that the Python interpreter directly executes the code without the need for a separate compilation step. This allows for an interactive programming environment where you can write and test code snippets quickly. The interpreter reads the source code, translates it into machine instructions, and executes it, providing immediate feedback.

Writing Simple Expressions

In Python, you can write expressions that perform calculations or evaluate conditions. An expression is a combination of values and operators that produces a result. For example:

- `3 + 7` evaluates to `10`.
- `3 < 15` evaluates to `True`.
- `'print me'` evaluates to the string itself.

You can also use the `print` function to display output on the console. For example:

- `print('Hello, world!')` will output `Hello, world!` to the console.

Arithmetic Operators

Python supports several arithmetic operators:

- `+` for addition
- `-` for subtraction
- `*` for multiplication
- `/` for division
- `%` for modulus (remainder)
- `**` for exponentiation

The order of operations, or precedence, determines how expressions are evaluated. For instance, multiplication and division have higher precedence than addition and subtraction. You can use

parentheses to enforce a specific order of evaluation:

- `(1 + 3) \* 4` evaluates to `16`.

Working with Integers and Real Numbers

Python can handle both integers and real numbers (floating-point numbers). When performing division with integers, the result is also an integer:

- `35 / 5` results in `7`.

- `84 / 10` results in `8`.

When you divide real numbers, the result is a floating-point number:

- `15.0 / 2.0` results in `7.5`.

Using Variables

A variable is a named piece of memory that can store a value. You can assign a value to a variable using the assignment statement:

- `x = 5`

- `gpa = 3.14`

Once a variable has been assigned a value, you can use it in expressions. For example, if `x = 7`, then `x + 4` evaluates to `11`.

Input and Output

You can read user input using the `input` function and display output using the `print` function. For example:

- `age = input('How old are you? ')`

- `print('Your age is', age)`

This will prompt the user for their age and then display it back to them.

Conclusion

In this lecture, we have introduced the basics of Python programming, including the definition of code, syntax, and the role of the Python interpreter. You have learned how to write simple expressions, use arithmetic operators, and work with variables. In the next lecture, we will delve deeper into control structures and data types in Python. Thank you for your attention, and I look forward to seeing you in the next session.

Quiz Questions

1. What is the role of the Python interpreter?

- To compile code before execution
- To execute code directly without compilation
- To write code for the programmer
- To store variables

Correct: To execute code directly without compilation

2. Which of the following is an arithmetic operator in Python?

- &&
- ||
- ++
- /

Correct: /

3. What does the print function do in Python?

- Stores a value in a variable
- Displays output on the console
- Reads user input
- Compiles the code

Correct: Displays output on the console